

Revisiting the prospects of photonic computing in the AI era

Programmable photonic meshes, in situ backpropagation, and what changed after 2022

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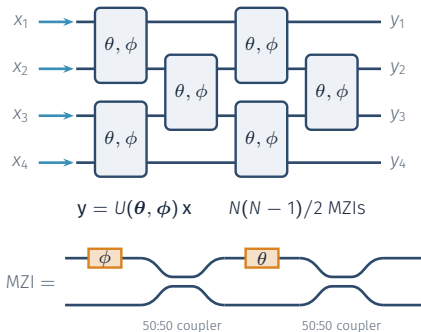
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PhD, Stanford University (2022)

The photonic mesh and its applications



Each box is a Mach-Zehnder interferometer: phase ϕ , a 50:50 coupler, phase θ , a second coupler. A rectangular mesh of them implements **any** $N \times N$ unitary.

Applications

- **Universal linear optics**
Reck 1994, Clements 2016
- **Self-configuring optics**
Miller 2013: mode sorting, unscrambling light, free-space links
- **Quantum photonics**
Carolan 2015: reconfigurable interferometers for photons
- **Neural networks**
Shen 2017: matrix multiplies for deep learning
- **Programmable photonics**
Bogaerts 2020: an FPGA for light

The 2010s thesis: photonic compute for quantum and AI

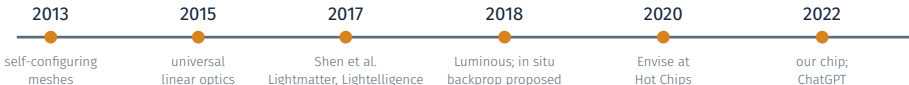
Quantum: the mesh *is* the computer

- Linear optics, single photons, and detectors are universal (KLM 2001)
- Reprogrammable 6-mode universal interferometer on a chip (Carolan et al., *Science* 2015)
- Boson sampling, then fusion- and measurement-based architectures built from switches and interferometers

AI: the mesh as a matrix engine

- Shen et al., *Nat. Photonics* 2017: coherent nanophotonic deep learning
- Premise: N^2 multiply-accumulates per pass for $\mathcal{O}(N)$ conversions and no data movement
- Training energy doubling every few months; a digital MAC costs ≈ 100 fJ

Well over \$1B of venture funding followed the AI thesis between 2017 and 2023.



The post-GPT thesis: photonic interconnects

Where the bottleneck went

- Transformer decode is memory-bandwidth bound: GPUs wait on HBM and on each other
- Models grew 100×+, clusters 10×, electrical I/O about 2×
- Co-packaged optics replaces pluggables: Nvidia photonic switches (2025), TSMC COUPE, Broadcom
- The AI datacenter adopted photonics for **moving** numbers, not multiplying them

The best-funded mesh company followed

- 2021: Passage interconnect announced beside Envisi
- 2024: \$400M Series D at \$4.4B, for interconnect
- 2025: M1000 ships in March; Envisi appears in *Nature* in April, as research

114 Tbps

Passage M1000: 4,000 mm², 256 fibers

\$850M

raised to date at a \$4.4B valuation

2%

of Envisi power is optical: 1.6 W of 79.6 W

Lightmatter press and blog, 2024–25; Ahmed et al., *Nature* 640, 368 (2025).

Optical neural network theory

A layer is a mesh plus a nonlinearity

$$\mathbf{x}^{(\ell+1)} = f\left(M^{(\ell)} \mathbf{x}^{(\ell)}\right), \quad M = V \Sigma U^\dagger$$

Any weight matrix is two unitaries and a diagonal of attenuators, so two meshes per layer suffice. One MZI is a 2×2 unitary $T(\theta, \phi)$; $N(N-1)/2$ of them in a grid reach all of $U(N)$ (Reck 1994; Clements 2016).

Where the energy goes, per multiply-accumulate

$$E_{\text{MAC}} \approx \underbrace{\frac{E_{\text{DAC}} + E_{\text{ADC}}}{N}}_{\text{conversions}} + \underbrace{\frac{2^{2b} \hbar \omega}{\eta N}}_{\text{photons, } b \text{ bits}} + \underbrace{\frac{P_{\text{hold}}}{f_{\text{clk}} N}}_{\text{holding phases}}$$

What the scaling says

- Every term falls as $1/N$: the mesh pays off only when it is **wide**
- Photons are cheap: 10^5 photons per 8-bit detection is well under a femtojoule per MAC
- Conversions are not: a GS/s 8-bit ADC costs picojoules, so N must reach the hundreds
- Footprint caps N near 64 to 256 per reticle, and error and loss compound over N stages

The bet

Optics wins if fan-in N is large, precision b is low, weights change slowly, and the data is already analog. In 2017, none of those described an AI workload.

In situ backpropagation and inverse design

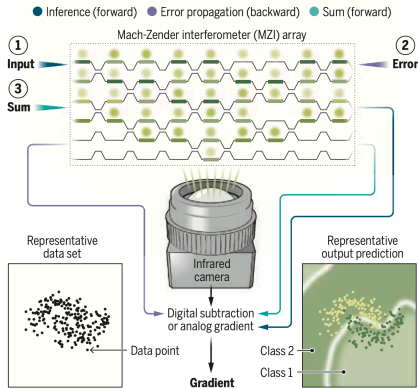


Figure: *Science* perspective on Pai et al., *Science* 380, 398 (2023).

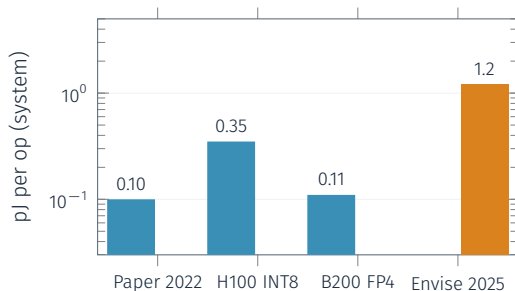
Same trick as photonic inverse design

- Adjoint method: a gradient is the interference of a forward and a backward field (Hughes, Minkov, Fan 2018)
- Forward input, backward error, then their sum: tap power at each phase shifter reads out $\partial\mathcal{L}/\partial\theta$

What the chip did

- 3-layer, 4-port silicon photonic network with thermal phase shifters and grating taps imaged by an IR camera
- **Analog update:** one toggle applies the gradient, with no digital subtraction and no $\mathcal{O}(N^2)$ readback
- Ring and moons trained *on chip*: 90 to 98% test accuracy, tracking digital training curves
- Simulated 64×64 MNIST: 97% ideal, 95% with tap noise and I/O error

Inference versus training: the energy budget



Digital bars: datasheet peak dense throughput over board power. Envisage: 65.5 TOPS at 78 W electrical plus 1.6 W optical (Harris, Apr 2025). Formats differ; read as orders of magnitude.

Reading the chart: a 2025 mesh system sits about 10× above GPU peak, with 98% of its power in electronics. The optics are not the term to optimize.

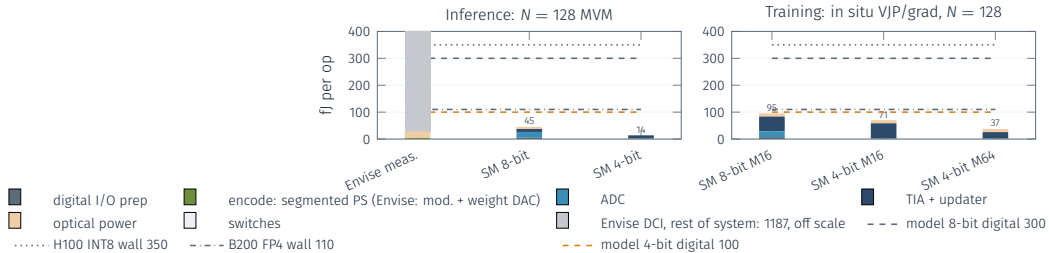
Inference: one forward pass

- Tolerates 4 to 8 bits, the analog-friendly regime
- Weights static, so phase holding amortizes
- Cost is dominated by **conversions**, not photons

Training: forward, backward, update

- Gradients want 16-bit dynamic range, analog's weak spot
- In situ backprop: the backward pass is one more optical pass, and the update is $\mathcal{O}(1)$ per phase instead of $\mathcal{O}(N^2)$ memory traffic
- But each step rewrites N^2 phases, so weight write energy and speed set the budget

Where the energy goes: the 2023 model, Enviser, and a 4-bit projection



2023 SM Tables S1 to S4, $N = 128$, 1 GHz, segmented phase shifters for inputs and weights (no DACs). Enviser: 78 W + 1.6 W optical at 65.5 TOPS, 0.25 W encode incl. weight DACs. 4-bit: $\text{ADC} \propto 2^b$, segments $\propto b$, digital op $\div 3$, inference light $\propto 4^b$, training light at the tap-SNR floor.

Reading the chart: segmented weights put the encode at 0.1 fJ/op, so the ADC dominates at 8 bits and the TIA at 4. Inference at 4 bits: $7\times$ under the 4-bit digital line; training $1.4\times$ ($M = 16$), $2.7\times$ ($M = 64$). The price: N^2b contacts, 65k at 4 bits, a bump array per weight.

What changed after 2022?

Precision collapsed

FP16 in 2017, FP8 with Hopper in 2022, FP4 with Blackwell in 2024. Inference at 4 bits is routine. The industry walked into the regime analog optics always needed.

Attention moved the bottleneck

Decode is KV-cache and weight-bandwidth bound. Softmax, normalization, and memory traffic are not matrix multiplies. A faster MVM engine attacks the wrong term.

Inference became the product

ChatGPT (Nov 2022), then reasoning models with test-time compute (2024). Inference now dominates deployed FLOPs and spend. Training chips are a market of a few buyers.

The ADC wall stayed put

Conversion energy per sample improves far slower than logic (Mурmann survey). Every photonic accelerator still pays it on every activation, in and out.

Revisit the four conditions: fan-in large (still hard), precision low (**now true**), weights static (**true for inference**), data already analog (not in a datacenter).

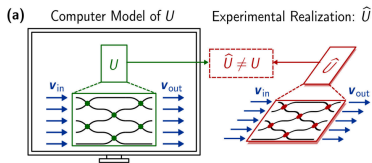
Other applications: quantum computing and analog signal processing

Quantum photonics: exactness first

- Meshes are the interferometers, switches, and multiplexers of linear-optical quantum computers (Carolan 2015; Xanadu and silicon platforms, *Nature* 2025)
- No DAC or ADC in the data path: photons in, detector clicks out
- Loss, extinction, and phase accuracy are what matter; accelerator calibration methods transfer directly

Analog signal processing and sensing

- Mode sorting and unscrambling for free-space and multimode links (Miller 2013; Annoni 2017)
- RF photonic filtering and beamforming, lidar wavefront control, optical switching
- The signal is already light: conversions only at the boundary, weights updated at kHz by feedback



The transferable toolkit: parallel programming of feedforward meshes, power monitoring with two detectors instead of taps, and in situ gradient measurement. Each was built to make an accelerator trustworthy; each is worth more where the mesh **is** the signal path.

Pai et al., JSTQE 2020; Nanophotonics 2023; Science 2023.

Questions?



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